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# A spruce beetle susceptibility map of Alaska using Sentinel-1 radar and crown-scale mortality data

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In Southcentral Alaska, an outbreak of the spruce beetle, Dendroctonus rufipennis (Kirby), has affected about 800 000 ha since 2016. During earlier outbreaks on the Kenai Peninsula and in the Copper River basin, the beetle preferentially attacked and killed large-diameter spruce of low radial growth, which were least able to resist colonization of the phloem. Outbreak hazard ratings score stand susceptibility from tree diameter, basal area, and the proportion of spruce. However, no field inventory has measured these stand metrics statewide. This study evaluated backscatter of 2014 to 2016, before the outbreak reached the Susitna valley, and a canopy height map against a crown-scale map of the spruce killed there by 2021. Every pre-outbreak scene over Alaska was composited into a 90 m map and tested against the aerial detection survey of forest damage. Killed stands were observed taller than surviving stands with 24.7 m against 12.9 m, and had returned more backscatter before the outbreak, -7.7 dB against -9.1 dB. Producer's accuracy, the share of a class classified correctly, was 91 per cent for killed and 76 per cent for surviving stands at the threshold that best separated them, and overall accuracy was 89 per cent. Against the survey, the map classified as susceptible 77 and 44 per cent of attacked and unattacked forest, respectively, on the Kenai Peninsula and 80 and 48 per cent in the Matanuska-Susitna Borough. In the Copper River basin, where attacked stands averaged 13.6 m and unattacked 10.4 m, less than 1 per cent of either reached the threshold and only canopy height separated them. Pre-outbreak backscatter therefore identified, without a field inventory, the tall stands the outbreak killed, and the map classified 29 per cent of Alaska's 906 521 km² of forest as susceptible stands that should be thinned before the next outbreak.

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## Introduction

The spruce beetle, Dendroctonus rufipennis (Kirby) [Coleoptera: Curculionidae: Scolytinae], is a bark beetle whose hosts include the spruces of North America, 'with preference for large-diameter, mature trees' (Fettig et al. 2026, p. 1). Its brood is semivoltine, maturing over two years, through most of its range, and univoltine, maturing in one, where the summer is warm enough for the larvae to develop to adults before winter (Hansen et al. 2001a). In Utah and Colorado and on the Kenai Peninsula, the share of univoltine brood was predicted best by the cumulative hours above 17 °C in the 40 to 90 days after peak flight (Hansen et al. 2001b), and a univoltine population grows and spreads at twice the rate of a semivoltine one (Raffa et al. 2008, p. 511). Between outbreaks, in its endemic phase, the beetle breeds in windthrown and felled spruce, which it prefers to standing trees (Schmid and Frye 1977, p. 11), and 'Once the availability of windthrown spruce is exhausted, mature adult spruce beetles often transition to attacking live spruces, which can lead to outbreaks' (Fettig et al. 2026, p. 1). A living spruce resists colonization with resin, and its 'defenses are overcome only when a critical minimum number of beetles are recruited to the host tree' (Bentz et al. 2010, p. 603), so an endemic population colonizes only weakened trees. 'Larger or thick-phloemed trees provide greater resources for larval feeding, and yield higher rates of beetle replacement' (Raffa et al. 2008, p. 512), and Raffa et al. (2008, p. 501) described bark beetle outbreaks as eruptions that 'occur when key thresholds are surpassed, prior constraints cease to exert influence, and positive feedbacks amplify across scales'. The large spruce is therefore the host in which the brood multiplies fastest and the one that only a mass attack can overcome, so it is colonized first when an outbreak begins, and the mean diameter of the attacked trees falls as the outbreak proceeds (Massey and Wygant 1954, Schmid and Frye 1976).

In Alaska, the beetle's preferred hosts are Lutz spruce (Picea × lutzii Little) and white spruce (P. glauca (Moench) Voss) (Fettig et al. 2026), and its outbreaks have been more frequent and severe in Lutz spruce than in white or Sitka spruce (P. sitchensis (Bong.) Carrière) (Holsten and Werner 1990). Outbreaks have recurred throughout the tree-ring record, which on the Kenai Peninsula showed regional outbreaks 'during the 1810s, 1850s, 1870–1880s, 1910s, and 1970–1980s', 'at a mean return interval of 52 years' (Berg et al. 2006, p. 219). Across south central and southwest Alaska, tree rings of 1601 to 2007 gave 'evidence of regional-scale outbreaks dating from the mid-1700s', and the outbreaks since the 1990s, though within that range, showed 'greater spatiotemporal synchrony' than at any time in about 250 years (Sherriff et al. 2011, p. 1459). In the Resurrection Creek watershed of the Kenai Peninsula, the beetle killed 29 per cent of the white spruce between 1976 and 1980, infesting the larger diameter classes first and the smaller trees as the larger were depleted (Werner and Holsten 1983), and across Alaska mortality was likewise concentrated in the larger spruce (Baker and Kemperman 1974, Holsten 1984), in the Copper River basin the more so the larger and the slower growing the tree (Doak 2004). On the Kenai Peninsula, attacked trees had grown more slowly than unattacked trees, and a large diameter predicted attack or death only when coupled with slow radial growth (Hard et al. 1983, Hard 1985). The Forest Service guide to those stands described the susceptible trees as 'often dominants or codominants in well-stocked stands' (Hard and Holsten 1985, p. 7), so the trees colonized first in an outbreak are often the tallest of the stand.

On the Kenai Peninsula, the timing of the outbreaks followed summer weather, in that the outbreaks of 1971 to 1996 'were positively associated with the 5-year backwards running average of summer temperature', and Berg et al. (2006, p. 219) suggested that warm temperature acted 'through a combination of increased overwinter survival, a doubling of the maturation rate from 2 years to 1

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year, and regional drought-induced stress of mature host trees'. Across south central and southwest Alaska, 'El Niño years, combined with severe late-summer drought, appeared to contribute significantly to spruce beetle outbreaks', which the cool phase of the Pacific Decadal Oscillation, a pattern of Pacific Ocean temperatures that shifts every few decades, tended to precede (Sherriff et al. 2011, p. 1459). In Utah and Colorado, outbreak years followed Decembers and preceding Septembers about 2 °C warmer than in other years and five years of below normal precipitation (Hebertson and Jenkins 2008, pp. 285–7), and in northwestern Colorado outbreaks were 'predicted best by interannual to multidecadal variability in drought, not by temperature alone' (Hart et al. 2014, p. 930). Warm summers raised beetle populations only where susceptible mature spruce remained, in that the Kenai relationship 'decoupled after 1996, presumably because spruce beetles had killed most of the susceptible mature spruce in the region' (Berg et al. 2006, p. 219). From 1920 to 1989 about 847 000 ha of Alaska's spruce forest were infested, and the outbreak of 1990 to 2000 alone infested 1.19 million ha, 'approximately 40% more forest area than was infested the previous 70 years', which Werner et al. (2006, p. 195) related 'to the relatively high densities of large-diameter spruce in an aging forest across the region and to a warming trend'. That outbreak peaked in 1996, when more than 500 000 ha of spruce stands were infested (Holsten and Hard 2001), and the present outbreak has affected about 2 million acres of Southcentral Alaska, some 800 000 ha, since 2016, and had spread north to Cantwell and the Alaska Range by 2020 (Zwieback et al. 2024b).

That host preference was translated into stand hazard ratings, in which a stand's susceptibility to spruce beetle attack is scored from the diameter, basal area, and spruce share of its trees (Schmid and Frye 1976, Reynolds and Holsten 1994a, Fettig et al. 2026). The first was Research Note RM-309 of the Rocky Mountain Forest and Range Experiment Station, in which Schmid and Frye (1976, p. 1) rated Engelmann spruce (Picea engelmannii Parry ex Engelm.) stands 'on the basis of physiographic location, tree diameter, basal area, and percentage of spruce in the canopy', the basal area being the cross-sectional area of the stems per hectare, and placed in the high hazard class the well-drained creek bottoms in which the spruce over 25 cm averaged more than 41 cm in diameter, the basal area exceeded 34 m² per hectare, and spruce formed more than 65 per cent of the canopy. Its diameter classes came from the White River outbreak of the 1940s in Colorado, in which the average diameter of the infested trees 'decreased from 21 to 17 to 15' inches as the large trees were depleted (Schmid and Frye 1976, p. 2). On the Kenai Peninsula, a decision tree on basal area, spruce share, the share of the spruce basal area in trees over 25 cm in diameter, elevation, and aspect predicted high hazard correctly for at least 73 per cent of the stands (Reynolds and Holsten 1994a). The current Forest Service leaflet rates as most susceptible the stands in drainage bottoms whose spruce average more than 40 cm in diameter, whose basal area exceeds 34 m² per hectare, and in which more than 60 per cent of the stand is live spruce (Fettig et al. 2026). A rating falls as an outbreak proceeds, so a stand is rated before the beetle reaches it, and in central Utah stands rated high to mostly medium hazard before the outbreak of 1986 to 1998 were rated medium to primarily low by 1998, after average spruce diameter, basal area, and overstory spruce had fallen (Dymerski et al. 2001). Every rating obtains its diameter, basal area, and spruce share from a field inventory, the ground measurement of the trees of a stand, so the preventive treatments a rating directs, thinning and partial cutting (Hard and Holsten 1985, Fettig et al. 2007, Hansen et al. 2010, Temperli et al. 2014), require the susceptible stands to be identified first.

Remote sensing of spruce beetle outbreaks has mapped mortality after it appeared rather than the stands susceptible before it. Landsat composites and dense time series mapped spruce beetle

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mortality in Colorado and its subtle change (Woodward et al. 2018, Ye et al. 2021), and the sensitivity of a Landsat time series to mortality depended on its severity (Rodman et al. 2021). In the Susitna valley, about 90 per cent of the dead spruce mapped at crown scale on WorldView-2 imagery of 2021 stood in stands where dead crowns covered no more than 5 per cent of the ground (Zwieback et al. 2024b, p. 412), and Landsat reflectance change related clearly to that cover only above 10 per cent (Zwieback et al. 2024b). Radar was scarcely used for insect damage before Sentinel-1 (Hollaus and Vreugdenhil 2019), and in Bavaria Sentinel-1 backscatter has since been used to predict the vitality of spruce attacked by bark beetles (Hechtl et al. 2025).

Radar backscatter and canopy height measure the structure of a stand. Backscatter, the part of the radar pulse that a stand reflects back to the satellite, rises with above-ground biomass, the dry mass of the trees, until it saturates, at about 100 tons per hectare at L-band (Dobson et al. 1992), and Sentinel-1 operates at the shorter C-band, where 'the C-band backscattering coefficient shows much less sensitivity to total above-ground biomass' (Dobson et al. 1992, p. 412). Canopy height is now mapped worldwide at 10 m (Lang et al. 2023), so a map of tall stands is a map of the large dominant spruce that the hazard ratings describe, made without a field inventory.

No study had scored radar backscatter or a canopy height map against the stands a spruce beetle outbreak killed, or mapped the stands susceptible to the beetle across a region. This study tested two hypotheses derived from that literature, and Table 1 lists the measures entered, the attribute of the hazard ratings each stands for, and the direction expected in killed stands. The first was that the pre-outbreak Sentinel-1 backscatter of a stand and its canopy height would rank the stands the outbreak killed above those that survived, because both measure the large dominant spruce the beetle colonizes first, and it was tested in the Susitna valley against the crown-scale map of dead spruce that Zwieback et al. (2024b) made from a WorldView-2 image of 2 August 2021. The second was that a statewide map of pre-outbreak backscatter would rank attacked above unattacked forest wherever the aerial detection survey, the yearly flights in which observers map forest damage from an aircraft, had recorded the outbreak, and where that test failed the map was to be narrowed to the regions in which it succeeded.

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**Table 1.** The measures the study entered, the attribute of the hazard ratings each stands for, the direction expected in the stands an outbreak killed, and the literature the expectation rests on.

| Measure | Indicator | Assumption | Basis |
|---|---|---|---|
| Canopy height before the outbreak, from three sources | Large dominant spruce | Taller | The beetle takes large, slowly growing dominants and codominants first (Massey and Wygant 1954, Werner and Holsten 1983, Hard and Holsten 1985, Doak 2004), and an outbreak left stands with smaller average and maximum heights (Veblen et al. 1991) |
| Sentinel-1 C-band VV level before the outbreak | Stand structure and biomass | Higher | Backscatter rises with forest biomass until it saturates, C-band being much less sensitive than L-band (Dobson et al. 1992, Santoro and Cartus 2018) |
| ALOS PALSAR L-band HV level | Stand structure and biomass | Higher | Backscatter rises with biomass to a saturation near 100 tons per hectare at L-band (Dobson et al. 1992) |
| Change in VV through the outbreak | Mortality | A difference of either sign, if the radar recorded the mortality | Radar was proposed for bark beetle damage, although detection before the crowns fade has had limited success (Hollaus and Vreugdenhil 2019, Hechtl et al. 2025) |
| Diameter and height of dead and live white spruce on inventory plots | Tree size | Dead trees larger and taller where the outbreak has run | Killed trees were larger and grew more slowly than survivors in the same stands (Werner and Holsten 1983, Hard 1985, Doak 2004) |
| Year the survey first recorded a stand | Order of attack | Taller stands recorded earlier | The largest trees died first and smaller ones as the outbreak proceeded (Massey and Wygant 1954, Schmid and Frye 1976, Werner and Holsten 1983) |

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## Methods

*Study area*

The study covered the forest of Alaska, 906 521 km² of 10 m cells with a canopy height of 5 m or more, which Sentinel-1 imaged from October 2014 (Figure 1). White and black spruce (P. mariana (Mill.) Britton et al.), Alaska birch (Betula neoalaskana Sarg.) and trembling aspen (Populus tremuloides Michx.) form the forest of the Interior and the Southcentral lowlands, with Lutz spruce dominant in the Kenai Peninsula and Sitka spruce along the coast.

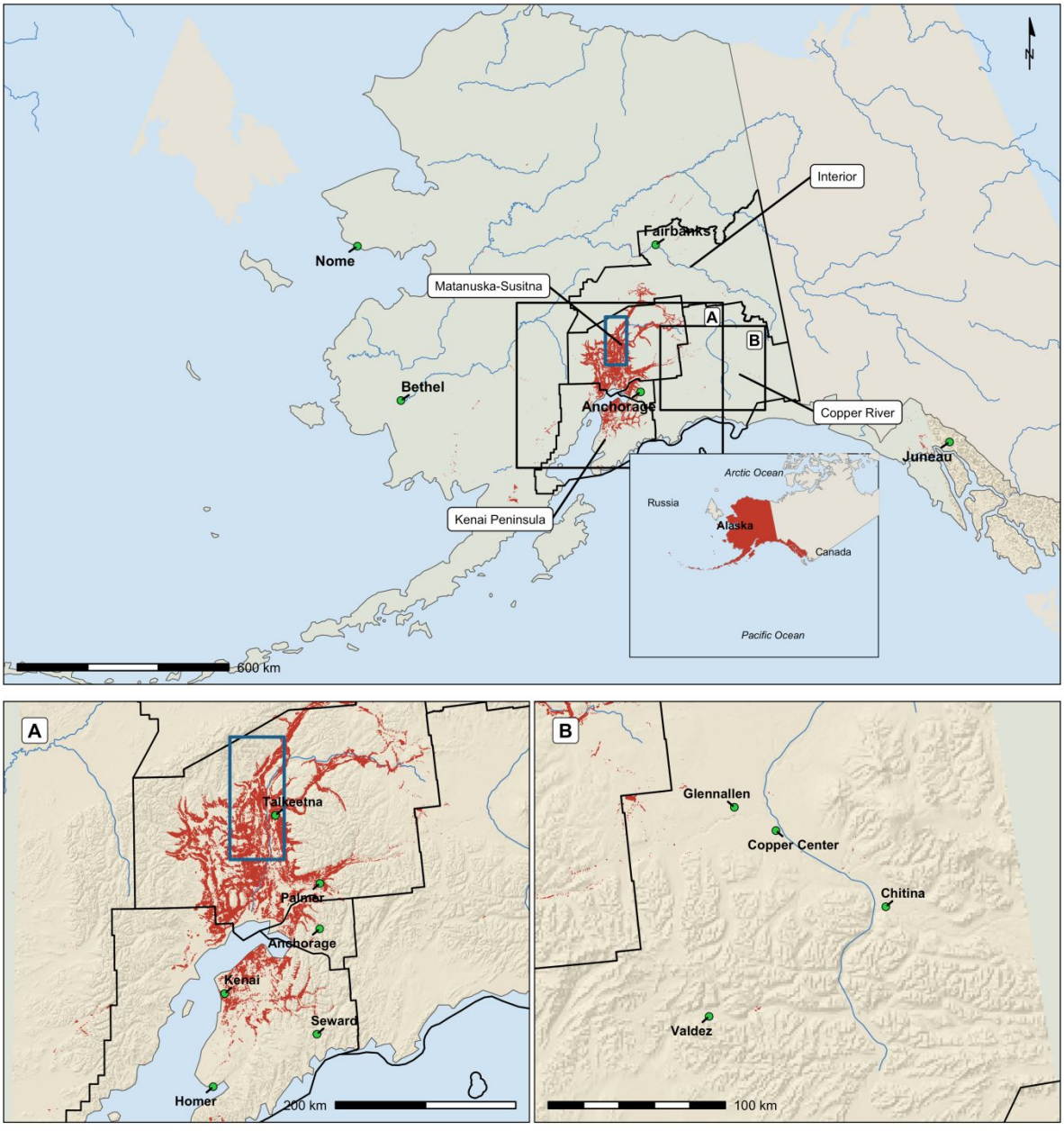



**Figure 1** The study area. Main map, Alaska in the Alaska Albers projection (EPSG:3338) over a shaded relief, with the spruce beetle damage mapped by the aerial detection survey from 2016 to 2024 in red, the four regions in which the statewide map was tested outlined in black, the validation strip of the Susitna valley in which Zwieback et al. (2024) mapped dead spruce

crowns on a WorldView-2 image of 2 August 2021 framed in blue, and the two detail panels boxed. A, Cook Inlet, the Kenai Peninsula and the Susitna valley. B, the Copper River basin. The inset places Alaska in North America.

The Susitna valley strip, 16.2 by 49 km along the Parks Highway between 62.1° N and 62.9° N, lies where the aerial detection survey mapped the most spruce beetle damage in 2018, 97 689 ha within 20 km of it. The four regions were the Kenai Peninsula Borough, which held most of the Alaska plot studies (Hard et al. 1983, Werner and Holsten 1983, Reynolds and Holsten 1994a), the Matanuska-Susitna Borough, which contains the strip, the Valdez-Cordova census area, which contains the Copper River basin of Doak (2004), and the Denali, Fairbanks North Star, and Southeast Fairbanks areas of the Interior, which the outbreak had reached at Cantwell by 2020 (Zwieback et al. 2024b).

*Reference data*

The dead spruce crowns that Zwieback et al. (2024b) delineated by hand on a WorldView-2 image of 2 August 2021 were the first reference dataset, 57 training areas of 512 m with 9 193 crowns and 121 test areas of 90 m with 3 147, with the 2 crowns of the areas delineated on a pre-outbreak image of 14 July 2013 set aside. The delineators were 'not able to directly identify a tree's species, whether it was alive or dead, or the mortality agent when assembling the test data' (Zwieback et al. 2024b, p. 418), so a crown is a dead spruce in an area of outbreak rather than a confirmed beetle kill, and it records that the spruce stood dead on that date and not the year it died. Each area was divided into 10 m cells on the WGS 84 UTM zone 5N grid of the labels, 159 182 cells in all, and the fraction of each covered by dead crowns was computed. From the same image, Zwieback et al. (2024b) published a severity map of the strip, the fraction of each 90 m cell covered by dead spruce, reported with a root mean square error of 0.02, which has no declared licence and was used for validation only. On the crown labels and on the severity map alike, a 90 m cell was classed dead at a dead crown fraction of 0.01 or more and living at a top cover of 5 per cent ormore, so that the living class held forest and not treeless ground, and that cover rule defined living forest in everetection survey of the ForestService, the yearly flights in which observers map forest damage from an aircraft, wthe second reference dataset e strip, its damage polygonsrecording survey year, host, and mortality class and its surveyed-area polygons the ground the observers flew oved Analysis plot and tree tablesfor Alaska (Bechtold and Patterson 2005) were the third, recording the diameter and height of every live or standclass for the dead, but no cause or year of death.

*Satellite data*

Sentinel-1 ground range detected scenes in interferometric wide swath mode were read in Google Earth Engine (Gorelick et al. 2017), where each had been corrected for thermal noise, calibrated, terrain corrected, and converted to decibels. Following the analysis-ready preparation of Mullissa et al. (2021), backscatter was normalized for slope with the volume scattering model of Vollrath et al. (2020), which also masks layover and radar shadow, on the Copernicus 30 m elevation model, because north of 60° N Mullissa et al. (2021) advised an alternative to the Shuttle Radar Topography Mission, and speckle was reduced with a 3 by 3 mean in linear power. Backscatter was read in VV and VH, the polarizations that send a vertical pulse and receive its vertical or horizontal return. The strip is imaged by the ascending orbit 65 and the descending orbit 131, which view the forest from opposite sides at different incidence angles, so the two were kept apart in the validation and pooled after a correction

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for incidence angle in the statewide map. Canopy height was read from the ETH Zurich model of 2020, which fitted the heights measured by the Global Ecosystem Dynamics Investigation lidar on the International Space Station to Sentinel-2 imagery at 10 m (Lang et al. 2023), and from two surface models less the 10 m terrain model of the United States Geological Survey 3D Elevation Program, the Copernicus GLO-30 surface from X-band radar of 2011 to 2015 and the ArcticDEM mosaic from optical stereo at 2 m (Porter et al. 2023). The yearly mosaics of the Phased Array L-band Synthetic Aperture Radar (PALSAR) on the Advanced Land Observing Satellite (ALOS) of the Japan Aerospace Exploration Agency gave L-band backscatter at 25 m in HH and HV, the horizontal pulse with its horizontal or vertical return (Shimada et al. 2014), the LANDFIRE existing vegetation type gave the vegetation class of each cell (Rollins 2009), and the conifer top cover of 2020 from the Arctic-Boreal Vulnerability Experiment (Macander et al. 2022) gave the cover rule above.

*Accuracy assessment*

Every test ranked dead against living cells or stands by one continuous source and scored the ranking by the area under the receiver operating characteristic curve, the chance that a dead unit drawn at random ranked above a living one (Equation 1), where R_i is the rank of dead unit i among all $n_d + n_l$ units of a test, with the 95 per cent confidence interval of DeLong et al. (1988).

$$\mathrm{AUC} = \frac{\sum_{i \in D} R_i - n_d(n_d + 1)/2}{n_d\, n_l} \tag{1}$$

The threshold that maximized the index of Youden (1950), the sum of the two producer's accuracies less one (Equation 2), where TP(τ) and TN(τ) are the dead and living units classified correctly at threshold τ, made each source a classifier that assumed no distribution of the backscatter, and at that threshold the confusion matrix gave the producer's accuracy of each class, the share of that class classified correctly, the commission and omission errors, Cohen's kappa (Cohen 1960), and the overall accuracy, the share of all labelled units classified correctly (Equation 3), with its exact binomial 95 per cent confidence interval.

$$J(\tau) = \frac{\mathrm{TP}(\tau)}{n_d} + \frac{\mathrm{TN}(\tau)}{n_l} - 1 \tag{2}$$

$$\mathrm{OA}(\tau) = \frac{\mathrm{TP}(\tau) + \mathrm{TN}(\tau)}{n_d + n_l} \tag{3}$$

Before any classifier, the separability of the two classes on each source was measured by the index M of Kaufman and Remer (1994), the difference of the class means $\mu_d$ and $\mu_l$ over the sum of their standard deviations σ_d and σ_l (Equation 4), with M above 1 taken as good separability, and tested with the Wilcoxon rank-sum test.

$$M = \frac{|\mu_d - \mu_l|}{\sigma_d + \sigma_l} \tag{4}$$

Where two sources were combined, logistic regressions, parametric binomial fits, were scored by the metric set of Murphy et al. (2026), Nagelkerke's $R^2$ (Nagelkerke 1991) with the likelihood ratio test,

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the Hosmer-Lemeshow test over ten groups (Hosmer and Lemesbow 1980), the Brier score (Brier 1950), the root mean square error, the same error relative to the mean of the label, the mean absolute error, Moran's I of the residuals over the eight nearest cells (Moran 1950), and the Durbin-Watson statistic of the residuals in cell order (Durbin and Watson 1950). Every fitted model was trained and scored over spatial folds dealt by blockCV (Valavi et al. 2018), so that no unit was scored by a model that had seen its neighbours, and overfitting was the root mean square error of the held-out folds over that of the training folds, rejected above 1.1 as in Murphy et al. (2026), with systematic bias measured as the bias share of Theil's U (Theil 1966) on the held-out predictions.

*Stand validation*

The first hypothesis was tested first on the stands of the label areas. For every month from October 2014 to August 2021, the month of the label image, the median slope-corrected backscatter of each orbit was composited in Earth Engine and sampled at the centre of every 10 m cell of every label area, VV and VH being available on both orbits from June 2017 and before then VV on both orbits with VH on the ascending orbit in scattered months from April 2015. Each 512 m area was divided into 90 m cells, the scale of the severity map, and the 1 543 cells wholly inside their area took the mean dead crown fraction and the mean backscatter, in linear power, of their 81 cells of 10 m. Because Zwieback et al. (2024b) related the change in Landsat shortwave infrared reflectance to severity, the change in VV between October 2014 to December 2016, before the survey recorded damage in any label area, and June 2019 to August 2021 was scored beside the level of each period, and the change was also tested with a Welch t-test, so that a difference that existed before the outbreak could be told from a change the outbreak made. To test whether the full monthly series added anything to the level, the temporal convolutional network of Pelletier et al. (2019), as implemented in sits (Simoes et al. 2021), was trained on the series of each stand over five spatial folds of 15 km hexagons across the 57 training areas, and its overfitting ratio used the predictions of each fold's network on its own training stands.

*Landscape validation*

Because the 1 543 stands were chosen by Zwieback et al. (2024b), the first hypothesis was then tested across the severity map. The mean slope-corrected VV of every scene of each orbit in the two periods, the three canopy heights, and the L-band mosaics of 2015 and 2016 were averaged into the 84 038 cells of the map with values, and each source was scored as above and by Spearman's rank correlation with severity over the forest cells. Logistic regressions on the cells with every source present tested whether a second source improved the ranking given by the first, over five folds in blocks of 1 km, no smaller than the 0.81 km autocorrelation range of the sources on the strip. The level of each source was also scored against the survey polygons of 2017 to 2024 rasterized onto the same cells, and the LANDFIRE class of each cell tested whether the height contrast was present within a cover class or was a contrast between host and non-host.

*Statewide map*

For For the second hypothesis, every Sentinel-1 VV scene over Alaska from October 2014 to December 2016 was pooled into one map. Slope-corrected VV falls with incidence angle across a swath, so the fall was fitted as the slope of VV on angle over the forest cells of 150 scenes drawn at random, at 300 m, and the median over the 131 scenes that covered forest, -0.125 dB per degree, brought every scene to the mid-swath angle of 38 degrees (Equation 5), where $\gamma^0(s,c)$ is the slope-corrected VV

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backscatter of scene $s$ at cell $c$ in decibels, $\theta(s,c)$ its local incidence angle and $\beta$ the fitted fall in decibels per degree.

$$\gamma^0_{38}(s,c) = \gamma^0(s,c) - \beta\,[\theta(s,c) - 38^\circ] \quad (5)$$

Forest was a 10 m cell with an ETH Zurich canopy height of 5 m or more. The product was computed at 90 m on the Alaska Albers grid, each scene read at the 90 m level of its image pyramid and the $n_c$ scenes over a cell averaged in linear power and returned to decibels as the pre-outbreak level (Equation 6), with the scene count, the mean canopy height, and the forest share of the 10 m cells as further bands, and was exported as the Earth Engine asset named in the code.

$$\bar{\gamma}^0(c) = 10\log_{10}\left[\frac{1}{n_c}\sum_{s=1}^{n_c} 10^{\gamma^0_{38}(s,c)/10}\right] \quad (6)$$

It was validated as made, read on the cells of the severity map and scored as above against the dead and living cells, the survey, and the ETH Zurich height, and its Youden threshold, -8.44 dB, defined a susceptible cell in everything that follows (Equation 7), the same rule giving 17.4 m on the height map.

$$S(c) = \begin{cases} 1 & \bar{\gamma}^0(c) \geq \tau \\ 0 & \bar{\gamma}^0(c) < \tau \end{cases} \quad (7)$$

*Regional validation*

The second hypothesis was then tested against the aerial detection survey in the four regions of Figure 1. In each region, a 90 m grid was laid on the Alaska Albers projection, and the survey's spruce beetle polygons on white spruce of 2016 to 2024 gave each cell its first year inside a polygon and the highest mortality class recorded over it. A cell was dead if it was living forest inside such a polygon, and living if it was living forest inside the ground flown in those years, had never lain inside a spruce beetle polygon of any host or year, and lay within 10 km of a white spruce polygon, so that the living class held forest the beetle could have reached and the surveyors had seen, and the cells of the severity map were excluded. In each region, 5 000 dead and 5 000 living cells were drawn at random with a fixed seed, or every cell where fewer existed, and scored by the area under the curve and by the share of each class at or above the strip's threshold. Spearman's rank correlation between the year a dead cell was first recorded and its canopy height tested whether the beetle moved from taller to shorter stands, as Werner and Holsten (1983) recorded on the Kenai Peninsula and Massey and Wygant (1954) on the White River. In the Matanuska-Susitna Borough, the nine single sources of the strip, with the PALSAR mosaics of 2007 to 2010 that predate every damage polygon, were also read at 20 000 dead and 20 000 living cells drawn the same way, and scored again with the dead cells restricted to polygons of 20 per cent mortality or more, to measure the ceiling the survey label set on any source.

*Ground plots*

The second hypothesis was checked on the ground against the Forest Inventory and Analysis plots, comparing the dead with the live white spruce of each plot as the plot studies compared killed and surviving trees (Werner and Holsten 1983, Doak 2004). Every forested plot of the four regions was taken at its latest measurement, 129 plots with white spruce in the Matanuska-Susitna Borough

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(measured 2018 to 2020), 262 on the Kenai Peninsula (1999 to 2025), 136 in the Valdez-Cordova census area (2002 to 2025) and 317 in the Interior with the Yukon-Koyukuk area added (2014 to 2019). The Interior plots were measured before the survey recorded the outbreak's front along the Nenana River, 99.8 per cent of whose cells it first recorded in 2020 or later, so they stood for white spruce the beetle had not yet reached. On the plots holding both live and dead white spruce, the mean diameter and height of the dead trees were set against those of the live trees with a paired Wilcoxon signed-rank test. Across all plots, the height of the tallest tree was correlated with the share of white spruce basal area dead and ranked the plots with at least half of that basal area dead by the area under the curve. The radar and height sources were read at each plot's published coordinate, which the inventory shifts by up to about 1.6 km, for the 90 m cell holding it and as the mean of the cells within 800 m.

## Results

### *Stand validation*

At the 90 m stand scale, the pre-outbreak VV level of October 2014 to December 2016 ranked the 369 stands the outbreak later killed above the 141 living stands with an area under the curve of 0.92 on the descending orbit and 0.89 on the ascending orbit (Table 2), the dead stands being 1.9 dB brighter before the beetle arrived. The change in that level to June 2019 to August 2021 did not separate the two classes, with an area under the curve of 0.50 to 0.55, because the dead stands fell by 0.20 dB and the living by 0.14 dB [Welch t-test, t(169) = -1.35, p = 0.178]. Conifer cover alone gave 0.55, so the pre-outbreak brightness did not measure canopy density. The temporal network trained on the full monthly series reached an F1 score of 0.92 and an area under the curve of 0.91 on the held-out folds, but its root mean square error was 0.12 on its own training stands against 0.28 on the held-out folds, a ratio of 2.33 against the 1.1 line of Murphy et al. (2026), so it was overfitted. Because the untrained pre-outbreak level matched it at 0.92, the level alone was adopted in landscape and statewide tests.

**Table 2.** Area under the receiver operating characteristic curve for ranking the stands the outbreak killed above the living stands, by source and by label. The crown labels are the 90 m stand cells of the Zwieback et al. (2024) training areas, the severity map their 90 m map of 2021, and the survey the aerial detection survey polygons of 2016 to 2024 across the rest of the Matanuska-Susitna Borough. The statewide product is the pooled, angle-corrected VV level of every scene; the two orbits are the single-orbit levels of the same months.

| Source | Crown labels | Severity map | Matanuska-Susitna Survey |
|---|---|---|---|
| Sentinel-1 VV, statewide product | | 0.86 | |
| Sentinel-1 VV, ascending orbit 65 | 0.89 | 0.88 | 0.63 |
| Sentinel-1 VV, descending orbit 131 | 0.92 | 0.80 | 0.61 |
| ETH canopy height 2020 | | 0.89 | 0.70 |
| GLO-30 surface minus terrain | | 0.84 | 0.64 |
| ArcticDEM surface minus terrain | | 0.84 | 0.50 |
| PALSAR L-band HV | | 0.86 | 0.72 |
| Conifer top cover | 0.55 | | |

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*Landscape validation*

On the severity map, 34 990 dead and 6 774 living forest cells among its 84 038 cells, the pre-outbreak VV level ranked dead above living with an area under the curve of 0.88 on the ascending orbit and 0.80 on the descending. Spearman's rho between the level and the dead crown fraction over every forest cell was 0.43 and 0.29 on the two orbits, respectively, and on the descending orbit the mean level rose with every severity bin, from -9.48 dB at zero to -7.76 dB at 0.1 or more, while the change between the two periods gave 0.48 and 0.62 on the same orbits. The three canopy heights ranked the cells as well as the radar did. The ETH Zurich height gave 0.89, with a mean of 24.7 m for the dead cells against 12.9 m for the living, and the GLO-30 and ArcticDEM surface minus terrain heights gave 0.84 each. L-band HV of 2015 and 2016 gave 0.86, and height and the C-band level were correlated over the forest cells [Spearman's rho 0.64]. On the 41 764 cells with every source, logistic fits gave an area under the curve of 0.887 for height alone, 0.884 for C-band alone and 0.918 for all three together. Within every LANDFIRE class with at least 100 cells of each kind, height ranked dead above living with areas under the claska Sub-boreal White SpruceForest, the one white spruce class with enough cells, height gave 0.86, with a me21.0 m for the dead cells aga, and logistic fits gave 0.808for the class alone and 0.904 for class and height together.

No single source reached a separability of M = 1 on the severity map, the index running from 0.54 for the GLO-30 height to 0.90 for the ETH Zurich height, though every source separated the two classes [Wilcoxon rank-sum, $p < 0.001$] (Table 3). At their Youden thresholds the sources reached an overall accuracy of 81 to 89 per cent and a Cohen's kappa of 0.42 to 0.61. The commission error of the dead class stayed below 7 per cent, while that of the smaller living class ran from 38 to 56 per cent. The logistic fits reached a Nagelkerke's $R^2$ of 0.27 to 0.59 and a Brier score of 0.064 to 0.105 (Table 4). Their root mean square error was 0.25 to 0.32 against a mean label of 0.84, 30.3 to 38.7 per cent of that mean, and their mean absolute error 0.13 to 0.21. None was overfitted, the ratio of held-out to training error being 1.000 to 1.002 and the Theil's U bias share 0.000 for every fit. The Hosmer-Lemeshow test rejected the calibration of every fit, and the residuals were clustered in space [Moran's I 0.34 to 0.53, $p < 0.001$; Durbin-Watson 1.07 to 1.42].

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**Table 3.** Separability and classification accuracy of each source at its own Youden threshold against the dead and living cells of the severity map, and of the temporal network at a probability of 0.5 on the held-out folds of the crown labels. M is the separability index of Kaufman and Remer (1994), the difference of the class means over the sum of their standard deviations, starred by the Wilcoxon rank-sum test; AUC is the area under the receiver operating characteristic curve with the 95 per cent confidence interval of DeLong et al. (1988); the threshold is in the source's own unit, decibels for radar, metres for height and probability for the network; κ is Cohen's kappa; commission and omission errors are given for each class; OA is the overall accuracy with its exact binomial 95 per cent confidence interval. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$, **** $p \le 0.0001$.

| Source | Dead | Living | M | AUC (95% CI) | Limit | κ | CA dead | CA living | OA dead | OA living | OA (95% CI) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sentinel-1 VV, statewide product | 34,990 | 6,774 | 0.61**** | 0.86 (0.86, 0.87) | -8.44 | 0.61 | 0.05 | 0.38 | 0.09 | 0.24 | 0.89 (0.88, 0.89) |
| Sentinel-1 VV, ascending orbit 65 | 34,990 | 6,774 | 0.81**** | 0.88 (0.87, 0.88) | -9.04 | 0.59 | 0.05 | 0.42 | 0.11 | 0.23 | 0.87 (0.87, 0.88) |
| Sentinel-1 VV, descending orbit 131 | 34,990 | 6,774 | 0.56**** | 0.80 (0.80, 0.81) | -8.84 | 0.42 | 0.07 | 0.56 | 0.17 | 0.30 | 0.81 (0.80, 0.81) |
| ETH canopy height 2020 | 34,990 | 6,774 | 0.90**** | 0.89 (0.88, 0.89) | 17.45 | 0.53 | 0.04 | 0.49 | 0.15 | 0.19 | 0.84 (0.84, 0.85) |
| GLO-30 surface minus terrain | 34,990 | 6,774 | 0.54**** | 0.84 (0.83, 0.84) | 0.42 | 0.50 | 0.05 | 0.52 | 0.16 | 0.22 | 0.83 (0.83, 0.83) |
| ArcticDEM surface minus terrain | 34,990 | 6,774 | 0.62**** | 0.84 (0.83, 0.85) | 12.20 | 0.47 | 0.06 | 0.52 | 0.15 | 0.29 | 0.83 (0.82, 0.83) |
| PALSAR L-band HV | 34,990 | 6,774 | 0.84**** | 0.86 (0.86, 0.87) | -14.08 | 0.56 | 0.05 | 0.44 | 0.11 | 0.25 | 0.86 (0.86, 0.87) |
| PALSAR L-band HH | 34,990 | 6,774 | 0.65**** | 0.82 (0.81, 0.83) | -8.68 | 0.53 | 0.06 | 0.45 | 0.11 | 0.32 | 0.86 (0.86, 0.86) |
| Temp network, blind crown folds | 369 | 141 | | 0.91 (0.88, 0.94) | 0.50 | 0.70 | 0.10 | 0.16 | 0.05 | 0.28 | 0.88 (0.85, 0.91) |

**Table 4.** Fit and overfitting diagnostics of the logistic fits of the dead and living cells of the severity map on each source and combination, and of the temporal network on the crown labels. $R^2$ is Nagelkerke's, starred by the likelihood ratio test against the null fit; HL is the Hosmer-Lemeshow statistic over ten groups, starred by its test; Brier is the mean squared error of the fitted probability against the label, RMSE its root, rRMSE the root relative to the mean of the label in per cent and MAE the mean absolute error; the RMSE ratio is the root mean square error of the held-out folds over that of the training folds across five spatial folds, 1.1 being the line above which a fit is overfitted (Murphy et al. 2026); U bias is the bias share of Theil's U on the held-out predictions; I is Moran's I of the residuals over the eight nearest cells, starred by its test; DW is the Durbin-Watson statistic of the residuals in cell order. * $p \le 0.05$, ** $p \le 0.01$, *** $p \le 0.001$, **** $p \le 0.0001$.

| Model | n | $R^2$ | HL | Brier | RMSE | rRMSE (%) | MAE | RMSE ratio | U bias | I | DW |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Sentinel-1 VV, statewide product | 41,764 | 0.423**** | 2074**** | 0.075 | 0.274 | 32.7 | 0.162 | 1.00 | 0.000 | 0.432**** | 1.26 |
| Sentinel-1 VV, ascending orbit 65 | 41,764 | 0.487**** | 641**** | 0.074 | 0.273 | 32.6 | 0.154 | 1.00 | 0.000 | 0.444**** | 1.24 |
| Sentinel-1 VV, descending orbit 131 | 41,764 | 0.267**** | 667**** | 0.105 | 0.324 | 38.7 | 0.213 | 1.00 | 0.000 | 0.476**** | 1.25 |
| ETH canopy height 2020 | 41,764 | 0.456**** | 212**** | 0.086 | 0.293 | 34.9 | 0.173 | 1.00 | 0.000 | 0.451**** | 1.21 |
| GLO-30 surface minus terrain | 41,764 | 0.274**** | 3198**** | 0.103 | 0.321 | 38.4 | 0.214 | 1.00 | 0.000 | 0.529**** | 1.07 |
| PALSAR L-band HV | 41,764 | 0.464**** | 369**** | 0.080 | 0.283 | 33.7 | 0.160 | 1.00 | 0.000 | 0.356**** | 1.38 |
| Two canopy heights | 41,764 | 0.465**** | 681**** | 0.082 | 0.286 | 34.2 | 0.169 | 1.00 | 0.000 | 0.455**** | 1.20 |
| Two C-band orbits | 41,764 | 0.517**** | 621**** | 0.069 | 0.263 | 31.4 | 0.144 | 1.00 | 0.000 | 0.427**** | 1.25 |
| L-band HV and HH | 41,764 | 0.474**** | 265**** | 0.080 | 0.282 | 33.7 | 0.159 | 1.00 | 0.000 | 0.336**** | 1.42 |
| C-band and heights | 41,764 | 0.559**** | 133**** | 0.068 | 0.261 | 31.1 | 0.139 | 1.00 | 0.000 | 0.425**** | 1.25 |
| L-band and heights | 41,764 | 0.539**** | 199**** | 0.072 | 0.269 | 32.1 | 0.146 | 1.00 | 0.000 | 0.349**** | 1.39 |
| All six sources | 41,764 | 0.594**** | 44**** | 0.064 | 0.253 | 30.3 | 0.130 | 1.00 | 0.000 | 0.365**** | 1.35 |
| Temp network, blind crown folds | 510 | | | 0.094 | 0.306 | 42.3 | 0.147 | 2.33 | 0.007 | 0.248**** | 1.41 |

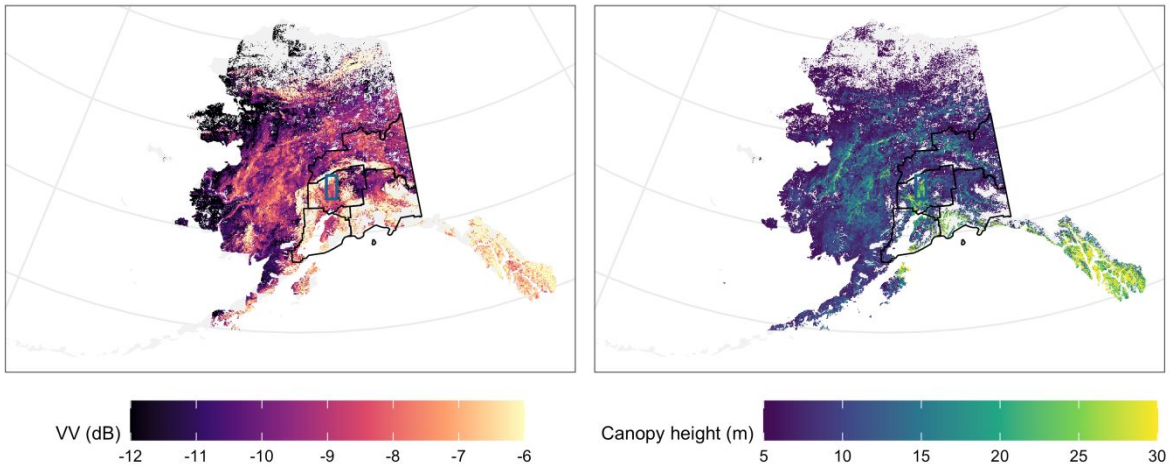



**Figure 2** The statewide susceptibility map at 1 km, the mean of the 90 m product over the forest cells of each 1 km cell. Left, the slope-corrected Sentinel-1 VV level of October 2014 to December 2016 brought to a 38 degree incidence angle; right, the ETH canopy height of 2020. Cells with less than a fifth of their area in forest are blank. Black outlines are the four test regions and the blue frame the validation strip.

*Statewide map*

Read on the 41 764 labelled cells of the severity map from a median of 57 scenes per cell, the statewide product ranked dead above living with an area under the curve of 0.86, the mean VV being -7.75 dB for the dead cells and -9.10 dB for the living. Its Spearman's rho with the ETH Zurich height over the forest cells was 0.71. At the threshold of -8.44 dB it classed 91 per cent of the dead cells and 24 per cent of the living cells as susceptible, an overall accuracy of 89 per cent. Against the aerial detection survey polygons on the same cells it gave 0.67. Across Alaska the map covered 906 521 km² of forest, 29 per cent of it at or above the threshold (Figure 2), and the susceptible share ranged from 27 per cent in the Interior to 62 per cent on the Kenai Peninsula (Table 5).

**Table 5.** Forest, susceptible forest and tall forest in the four test regions and in Alaska, from the 1 km map. Forest is the area of 10 m cells with a canopy height of 5 m or more. Susceptible forest is the forest whose pre-outbreak VV level reached -8.44 dB, the threshold that best separated the dead from the living cells of the severity map, and tall forest is the forest whose canopy height reached 17.45 m, the threshold the canopy height map took on the same cells. Scenes is the median number of Sentinel-1 scenes behind a forest cell.

| Region | Forest (km²) | Susceptible (km²) | Susceptible (%) | Tall stands (km²) | Scenes |
|---|---|---|---|---|---|
| Kenai Peninsula | 25 856 | 16 139 | 62 | 6 366 | 49 |
| Matanuska-Susitna | 47 011 | 22 617 | 48 | 9 976 | 57 |
| Copper River | 51 186 | 25 754 | 50 | 7 439 | 63 |
| Interior | 81 395 | 21 885 | 27 | 4 174 | 72 |
| Alaska | 906 521 | 258 717 | 29 | 106 336 | 53 |

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*Regional validation*

Against the aerial detection survey, the radar map ranked attacked above unattacked forest with an area under the curve of 0.70 on the Kenai Peninsula and 0.70 in the Matanuska-Susitna Borough, where the attacked stands were 0.8 dB brighter and 5.7 m taller than the unattacked (Table 6). In the Copper River basin, where the survey had recorded damage only since 2022, it gave 0.44, and both classes were darker, -10.1 against -10.0 dB, and shorter, 13.6 against 10.4 m, than in the two southern regions. In the Interior it gave 0.61, where the 5 000 dead cells all lay between 63.3° N and 63.8° N along the Nenana River and 98 per cent of them were first recorded in 2021 or later. The height map gave 0.67, 0.70, 0.76 and 0.58 in the same four regions, respectively. In the Matanuska-Susitna Borough the nine single sources scored 0.50 to 0.72 against the same label, the highest being L-band HV of 2007 to 2010, and limiting the dead cells to polygons of 20 per cent mortality or more changed nothing, 0.51 to 0.70, so the survey label and not the source limited the score. The stands the survey first recorded in 2016 and 2017 were the tallest, 24.9 and 24.8 m on the ETH Zurich height, and those first recorded in 2024 the shortest, 19.8 m, with a Spearman's rho between year and height of -0.23 over cohorts of 645 to 4 417 cells.

*Ground plots*

On the 77 Matanuska-Susitna plots holding both live and dead white spruce, the dead trees were the larger, averaging 24.2 cm in diameter and 14.6 m in height against 15.7 cm and 10.4 m for the live trees [paired Wilcoxon signed-rank test, $V = 2\,811$, $p < 0.001$ for diameter and $V = 2\,779$, $p < 0.001$ for height]. Across the 129 plots, the share of white spruce basal area standing dead rose with the height of the tallest tree [Spearman's rho 0.60], and that height ranked the 75 plots with at least half their basal area dead with an area under the curve of 0.84. On the Kenai Peninsula, the dead trees on the 173 plots holding both were again the larger, 22.0 cm and 13.2 m against 17.8 cm and 11.2 m for the live trees [$V = 12\,213$, $p < 0.001$ and $V = 11\,673$, $p < 0.001$]. In the Copper River basin the dead trees were thicker, 19.3 cm against 16.8 cm on 69 plots [$V = 1\,566$, $p = 0.032$], but no taller [$V = 1\,404.5$, $p = 0.240$]. On the 110 Interior plots holding both, the dead white spruce were no thicker than the live, 17.3 against 17.5 cm [$V = 2\,557.5$, $p = 0.184$], and were shorter, 11.3 against 12.0 m [$V = 2\,079.5$, $p = 0.004$]. The ETH Zurich canopy height around a plot's published coordinate ranked the half-dead plots of the Matanuska-Susitna Borough at 0.74, against 0.84 for the plot's own tallest tree.

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**Table 6.** The statewide product against the aerial detection survey in four regions. Dead cells lay inside a spruce beetle polygon on white spruce of 2016 to 2024; living cells were forest the surveyors flew over, never inside a spruce beetle polygon of any year, within 10 km of one. AUC is the area under the curve for ranking dead above living; the two shares are the cells of each class the strip's VV threshold called susceptible; rho is Spearman's correlation between the year a dead cell was first recorded and its canopy height.

| Region | Dead | Living | AUC, VV | AUC, height | VV dead (dB) | VV living (dB) | Susceptible dead (%) | Susceptible living (%) | rho year, height |
|---|---|---|---|---|---|---|---|---|---|
| Kenai Peninsula | 5 000 | 5 000 | 0.70 | 0.67 | -8.1 | -8.7 | 77 | 44 | 0.09 |
| Matanuska-Susitna | 5 000 | 5 000 | 0.70 | 0.70 | -7.9 | -8.7 | 80 | 48 | -0.23 |
| Copper River | 1 719 | 5 000 | 0.44 | 0.76 | -10.1 | -10.0 | 1 | 0 | 0.23 |
| Interior | 5 000 | 2 800 | 0.61 | 0.58 | -8.6 | -8.9 | 40 | 27 | -0.02 |

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## Discussion

In the Susitna valley, on the label areas of Zwieback et al. (2024b), the Sentinel-1 VV level of October 2014 to December 2016 ranked the stands the outbreak later killed above the stands that survived, while the change in the same cells through the outbreak ranked nothing. The first hypothesis was therefore supported against the only crown-scale reference data this outbreak has, and what the level measured was the structure of the stand before the outbreak rather than the mortality that followed. The product's Spearman's rho with the canopy height map over the forest was 0.71, so the pre-outbreak level measured stand height and the map is a map of tall stands, and the canopy height map, the C-band level and L-band HV each ranked the cells in the direction the literature review expected.

### *Stand height*

On the Kenai Peninsula, 'Mortality was greatest in the larger diameter classes during the early part of the infestation' (Werner and Holsten 1983, p. 96), in the Copper River basin, 'Mortality increased with increasing DBH and decreasing BAI' (Doak 2004, p. 810), that is with larger diameter and slower radial growth, and the Forest Service synthesis reduced those plot studies to the rule that 'Spruce beetles preferentially attack and kill large diameter trees that have been growing slowly at breast height' (Hard and Holsten 1985, p. 7). The present study found the same host selection at the scale of the stand, from radar and canopy height data with no beetle observation among the predictors, and the inventory plots confirmed it on the ground in the same decade. In the Matanuska-Susitna Borough, the dead white spruce stood 8.5 cm thicker and 4.2 m taller than the live trees beside them, while on the Interior plots, measured before the outbreak's front arrived, the dead trees were no larger than the live, so the size effect belonged to the beetle. The qualification of Hard et al. (1983, p. 678), that diameter mattered 'unless large diameter was coupled with slower than average radial growth in the last 5 years', is not in the map, because none of its sources records radial growth, and height alone ranked the strip's stands at 0.89 without it. The order of attack was the one Schmid and Frye (1976, p. 2) described on the White River, in that the stands the survey first recorded in the Matanuska-Susitna Borough, in 2016 and 2017, were the tallest and each later year's stands were shorter through to 2024. The contrast held inside each LANDFIRE class with enough cells, so it separated tall from short stands of the same cover type and not host from non-host.

### *Hazard rating*

The second hypothesis applied that finding to the stand hazard rating, which since Schmid and Frye (1976, p. 1) has scored stands 'on the basis of physiographic location, tree diameter, basal area, and percentage of spruce in the canopy'. The ratings take diameter, basal area and spruce share from a field inventory, and the thresholds now in use (Fettig et al. 2026, p. 8) have never been applied across the forest of a state. The map gave the stand structure part of such a rating at 90 m across 906 521 km² of forest from freely available data. Of that forest, 29 per cent lay at or above the threshold that separated the killed strip stands from the surviving, from 27 per cent in the Interior to 62 per cent on the Kenai Peninsula (Table 5). The map rated hazard and not risk, in the distinction Reynolds and Holsten (1994b) drew when a panel of experts set stand hazard and windthrown trees as 'the two most important factors determining risk of a spruce beetle outbreak', because it included no windthrow, no beetle population and no spruce share. A tall birch or aspen stand therefore scored as a tall spruce stand would, so a manager reads the map beside a stand type map. The treatment it

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directs is the one Hard and Holsten (1985, p. 2) prescribed before the beetle arrives, 'Mark all spr) or larger in diameter forremoval regardless of growth rate'.

*Aerial survey*

Against the aerial detection survey, the map ranked attacked above unattacked forest at 0.70 on the Kenai Peninsula and 0.70 in the Matanuska-Susitna Borough. In the borough every one of nine sources scored between 0.50 and 0.72 against the survey label, where the crown-scale labels had scored them at 0.88 to 0.89. The survey is sketch-mapped from an aircraft at a scale at which a polygon of 2 per cent mortality is mostly living forest, and restricting the dead cells to polygons of 20 per cent mortality or more changed nothing, so the survey label and not the map limited the regional scores, which are a lower bound on the map's skill. In the Copper River basin the radar level failed where the canopy height map did not, 0.44 against 0.76, because the attacked stands there averaged 13.6 m against 22.1 m in the Matanuska-Susitna Borough and both classes returned near -10 dB. The radar map was therefore validated for the Kenai Peninsula and the Susitna lowlands, and the height map is the product for the Copper River basin, the narrowing of extent the design allowed for. In the Interior the map scored 0.61 at the front of the outbreak along the Nenana River, where 98 per cent of the recorded cells date from 2021 or later and the polygons hold stands the beetle has entered but not yet killed.

*Limitations*

The crown labels and severity map are one image of one valley on one date, and the threshold was set on that valley, so a region whose stands are taller or shorter will have more or fewer cells above it than the beetle will attack. The canopy height map is a 2020 model whose estimates 'saturate around 25 to 30 m' (Lang et al. 2023, p. 1779), just above the mean of 24.7 m for the strip's dead stands, and it was trained on lidar footprints that reach no farther north than 51.6° N, so in Alaska it predicts beyond its training data. The GLO-30 and ArcticDEM surfaces, the PALSAR mosaic of 2007 to 2010 and the radar of 2014 to 2016 all predate the outbreak and gave the same ranking. The pooling of orbits depends on one fitted fall of backscatter with incidence angle, -0.125 dB per degree, and on 49 to 72 scenes behind a forest cell in the four regions. Crown-scale detection in Alaska has needed airborne hyperspectral and lidar data to reach '78% accuracy for all three health statuses' (Cessna et al. 2021), and Landsat detected the outbreak only above a severity of 0.1 (Zwieback et al. 2024b), whereas the map here detects no outbreak and only identifies, before one arrives, the stands it would attack first.

## Conclusion

Validated against the crown labels and severity map of Zwieback et al. (2024b), the pre-outbreak Sentinel-1 VV level and three independent canopy heights ranked the stands the spruce beetle killed in the Susitna valley above those that survived, the host selection the plot studies have reported since 1954 at the scale of the tree, found here at the scale of the stand with no beetle observation among the predictors. Pooled across every scene over Alaska, the level gave a 90 m map of susceptible stands that ranked attacked above unattacked forest on the Kenai Peninsula and in the Susitna lowlands, where the inventory plots confirmed the size effect on the ground, and that failed in the open woodland of the Copper River basin, where the canopy height map ranked them instead. The two maps give the stand structure part of a hazard rating without a field inventory and show forest managers which stands to thin first before the next outbreak.

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## Conflict of interest statement

None declared.

## Data availability statement

The data underlying this article are available in Google Earth Engine as the public assets projects/vvb-checks/assets/spruce_susceptibility_alaska_90m, the 90 m map of pre-outbreak backscatter with its scene count, canopy height, and forest share bands, and projects/vvb-checks/assets/spruce_susceptibility_alaska_90m_class, its susceptibility classes, and in the article and in its online supplementary material. The datasets were derived from sources in the public domain, which were the dead spruce crown labels of Zwieback et al. (2024b) in Zenodo at https://doi.org/10.5281/zenodo.10569990 (Zwieback et al. 2024a); the spruce beetle outbreak severity map of Southcentral Alaska in Zenodo at https://doi.org/10.5281/zenodo.8423568 (Zwieback 2023), which carries no declared licence and was used for validation only; the aerial detection survey damage polygons of the United States Forest Service, Alaska Region, at https://www.fs.usda.gov/foresthealth/docs/IDS_Data_for_Download/AK_Region10_AllYears.gdb.zip; the Forest Inventory and Analysis plot and tree tables for Alaska at https://apps.fs.usda.gov/fia/datamart/; the conifer top cover of Macander et al. (2022) in the Oak Ridge National Laboratory Distributed Active Archive Center at https://doi.org/10.3334/ORNLDAAC/2032; the canopy height map of Lang et al. (2023) at https://gee-community-catalog.org/projects/canopy/; and the Sentinel-1 ground range detected scenes (COPERNICUS/S1_GRD), the Copernicus GLO-30 digital elevation model, the ALOS PALSAR yearly mosaics, the ArcticDEM and 3DEP elevation models, the LANDFIRE existing vegetation type, and the TIGER boundary files, all read in Google Earth Engine under those catalogue names. The sampled cells, the screening table of the literature review, and the tables and figures are held with the manuscript, and every number in the text is computed from them at render time.

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